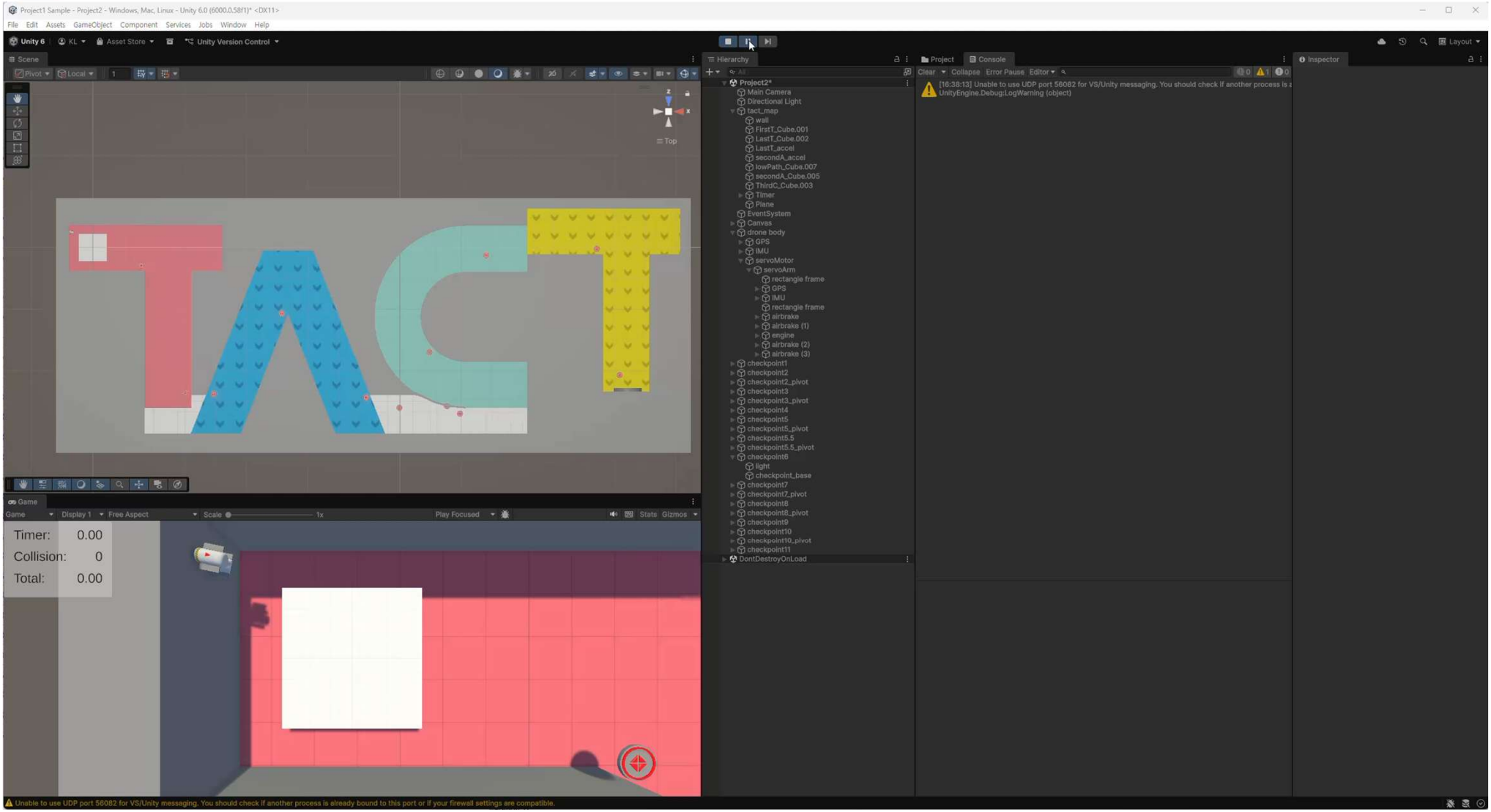
# TA session 07

Project 3 – LIDAR, SILO, Camera and UI

# Best Performance (10.54s)

Team 2

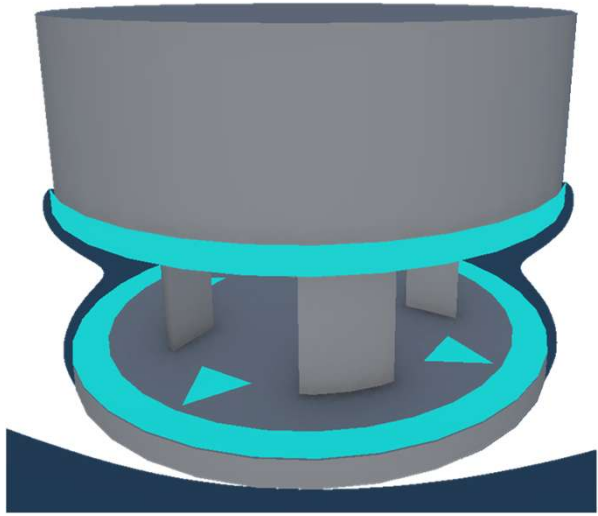


# Project3

- **Shared Autonomy**
  - Manual control + Automatic control
    - Can use 1 mothership with silo, and 3 drones
    - Can use everything we've learned before
  - New track
    - (To be announced next week)
  - Same rule
    - Collision will add +5s penalty time
      - Including every drones and mothership
    - fastest team will get +5 bonus for final score
  - Same parts
    - Roket body, Drone body and frames
    - engine, airbrake, servo motor
    - IMU, GPS

# New rules

- **New parts**
  - **LIDAR**
    - LIDAR_behave.GetDistance(float angleDegree)
  - **DroneSilo**
    - DroneSilo_behave.SpawnDrone(0)
    - DroneSilo_behave.RetreiveDrone()
- **New rules**
  - Random Gravity may appear during the track
  - Consider rotational stabilization or control



# Project3

1. **Design a mothership (Max thrust 60)**
   - Mothership must use "body" from project 1
   - Mothership can use maximum 3 drone silos (don't stack it)
2. **Design drones (Max thrust 30)**
   - Drones must use "drone body" from project 2
3. **Design manual + automatic control system**
   - You may manually control drone or automatically control mothership too
   - Consider using camera control and UI text
4. **Design controller**
   - 2 Linear potentiometer provided per team
   - **Don't forget to return every electronic parts provided for this project after presentation!**

# Project3

- **UI Text**: TMP_Text
  - TMP_Text text.text = string value
- **Draw debug line**
  - Debug.DrawRay(Vector3 origin, Vector3 size, Color color);
- **Camera Control**
  - CameraBehave.target = (object you want to follow)

# Restrictions

**If any of the restrictions are violated, the performance report will be disqualified. (0 points).**

- **Overall**
  - Do not drive off the given track (through a wall or across the y-axis)
  - If you want to use external libraries or assets, be sure to get prior approval.
  - You must not modify the structure, size, or components of the track or time, physics system in Unity.
- **Spacecraft**
  - Total size must fit in 9 * 9 * 5m box. **(both drone and mothership)**
  - Only one **body** part can be used in mothership, and it must be the highest parent in the hierarchy.
  - **You can only use 3 drone silo for mothership, and you must not stack them in y axis**
  - Only one **drone body** part can be used in each drone, and it must be the highest parent in the hierarchy.
  - Do not manipulate the size, weight or performance of the parts. (except frames)
  - The total thrust of all engines used in a rocket must be less than **30 for drone, 60 for mothership**.
  - A completed drone must not have any disconnected parts.
  - Except for the frame, no parts can overlap. **(servoarm will count as a frame)**
    - Frame-frame (o)
    - Frame-part (o)
    - Part-part (x)
  - Design so that interference does not occur even when the part is rotated by a servo motor.
- **Control**
  - Drone must run without any control except for start.

# Evaluation criteria

- **To be announced next week**